ON-BOARD DIAGNOSTIC [ENTERTAINMENT SYSTEM (WITHOUT CENTER DISPLAY)]

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Note
- "iPod" is a registered trademark of Apple Inc.

Outline
- The entertainment system has an on-board diagnostic function and diagnostic assist function to facilitate system diagnosis.

On-board Diagnostic Function
Malfunction detection function
- The on-board diagnostic function consists of the following functions:
  — A malfunction detection function, which detects overall malfunctions in the audio system-related parts.
  — A memory function, which stores detected DTCs.
  — A display function, which indicates system malfunctions via a DTC display.
- Detects a malfunction in the entertainment system.

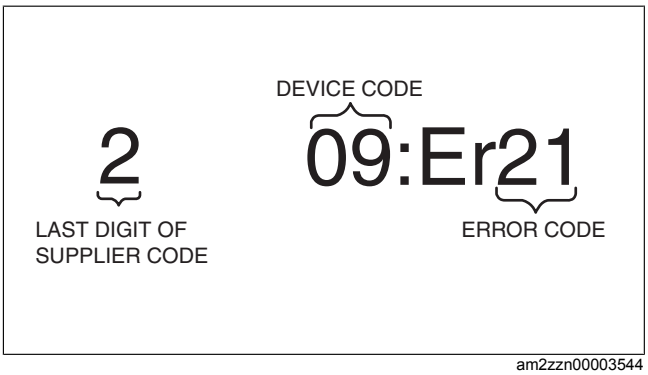
Memory function
- If the audio unit detects a malfunction in the entertainment system, it stores the DTC in the audio unit.

Target	Maximum memory number
Audio unit	3
CD player	3
Bluetooth® unit	0*

* : Because the Bluetooth® unit only displays the currently detected DTC, the memory function is not available.

- If the audio unit has already stored the maximum number of DTCs and detects another malfunction, it clears the oldest DTC and stores the new DTC.
- DTCs stored in the audio unit are cleared if the negative battery cable is disconnected (if backup power is interrupted).

Display Function
- To display DTCs in the audio unit display that have been recorded in the memory function, activate the on-board diagnostic function by operating the audio unit. Refer to the Workshop Manual for the activation procedure of the on-board diagnostic function.
  The DTCs consist of the following:
  — Supplier code (supplier name)
  — Device code (malfunction location)
  — Error code (malfunction contents)



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Supplier code	Supplier name
01	SANYO Automedia
02	Panasonic
03	Clarion
04	Pioneer
05	VISTEON
06	DESAY

Device code	Device name
09	Audio unit
16	CAN system
26	Bluetooth® unit
28	USB
29	iPod
30	CD player

Error code	Malfunction contents
01	Mechanical malfunction
07	Poor data read-outs
09	No playable File/Track
11	CAN communication time-out
12	CAN line (communication line) error

Error code	Malfunction contents
20	Insufficient power supply
21	Amplifier related circuit
30	Temperature abnormality

DTC table

For Audio Unit

×: Applicable
—: Not applicable

DTC No.	Warning/ indicator light	Description	Fail-safe	Drive cycle	Memory function
16:Er11*1	—	CAN communication time-out	—	—	×
16:Er12*1	—	Unit communication error	—	—	×
28:Er09	—	No USB playback function file/track	—	—	×
No Err	—	DTCs are not stored.	—	—	—

*1 : With CD player or with Bluetooth® unit

For Bluetooth® Unit

—: Not applicable

DTC No.	Warning/ indicator light	Description	Fail-safe	Drive cycle	Memory function
26:Er81	—	Communication error with Bluetooth® unit	—	—	—
26:Er82	—	Microphone circuit malfunction	—	—	—
26:Er83	—	Microphone circuit malfunction	—	—	—
26:Er84	—	Microphone circuit malfunction	—	—	—
26:Er85	—	Steering switch circuit malfunction	—	—	—
26:Er86	—	Bluetooth® unit internal malfunction	—	—	—
No Err	—	DTCs are not stored.	—	—	—

For CD Player

×: Applicable
—: Not applicable

DTC No.	Warning/ indicator light	Description	Fail-safe	Drive cycle	Memory function
30:Er01	—	CD player mechanism malfunction	—	—	×
30:Er07	—	CD player data read error	—	—	×
30:Er30	—	CD player temperature abnormality	—	—	×
No Err	—	DTCs are not stored.	—	—	—

Detection condition for the applicable DTC

For Audio Unit

DTC No.	Description	Detection condition
16:Er11	CAN communication time-out	• Time-out occurs in the audio unit CAN signal.
16:Er12	Unit communication error	• The audio unit cannot receive CAN signal.
28:Er09	No USB playback function file/track	• Time-out occurs in the audio unit CAN signal.

For Bluetooth® Unit

DTC No.	Description	Detection condition
26:Er81	Communication error with Bluetooth® unit	• The audio unit cannot receive CAN signal from the Bluetooth® unit.

DTC No.	Description	Detection condition
26:Er82	Microphone circuit malfunction	• Short to power supply detected in microphone input circuit.
26:Er83	Microphone circuit malfunction	• Short to power ground detected in microphone power supply circuit.
26:Er84	Microphone circuit malfunction	• Microphone non-connection signal detected continuously.
26:Er85	Steering switch circuit malfunction	• Steering switch input signal is detected continuously for **2 min or more**.
26:Er86	Bluetooth® unit internal malfunction	• Bluetooth® unit detects the internal malfunction.

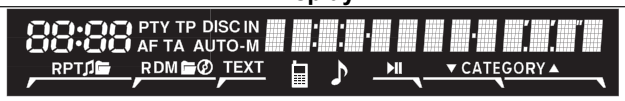
For CD Player

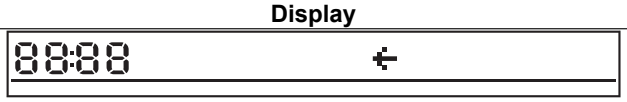
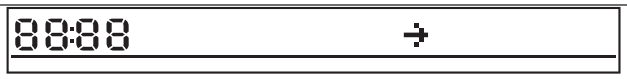
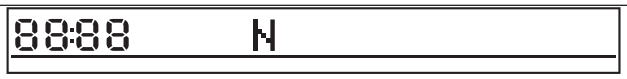
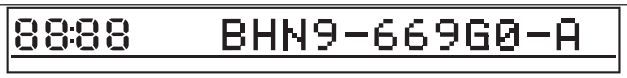
DTC No.	Description	Detection condition
30:Er01	CD player mechanism malfunction	• CD player mechanism malfunction detected.
30:Er07	CD player data read error	• CD player read error detected.
30:Er30	CD player temperature abnormality	• Abnormal temperature in CD player is detected.

Diagnostic Assist Function

- The entertainment system related parts are forcibly operated to enable verification of the connection condition and operation status of the entertainment system related parts. In addition, the specifications for the entertainment system related parts are displayed in the audio unit display to enable their verification. Refer to [DIAGNOSTIC ASSIST FUNCTION [AUDIO]] in the workshop manual for the diagnostic assist function activation procedure.

Diagnostic assist function table

Display	Function	Content
	LCD inspection	Displays all characters in the audio unit display.
—	Button inspection	Sounds buzzer when the audio unit button is operated.
—	Speaker inspection	Outputs sound from the speakers in the following order: 1. Front door speaker (LH) 2. Front door speaker (RH) 3. Rear door speaker (RH) 4. Rear door speaker (LH)
	Radio reception condition inspection	Displays the radio reception status in the audio unit display.
	Antenna control inspection	Displays the power supply output status to the AM/FM antenna amplifier in the information display.
		

Display	Function	Content
	Dial inspection	Displays the rotation direction of the operated dial switch in the audio unit display.
		
	Audio amplifier specification verification	Displays the audio amplifier specification in the audio unit display.
CD PLAYER IS NORMAL (INSERTED CD IS NOT COMPATIBLE) 	CD inspection	Displays the result of the CD inspection in the audio unit display. **Note** • If the CD inspection cannot be performed normally, TEST REJECTED is displayed in the audio unit display and the CD inspection mode stops.
MALFUNCTION IN CD PLAYER 		
—	Hands-free password clearing	Clears the Hands-free password.
	CD part number verification	Displays the CD player part number in the audio unit display.